Advanced Quantum Mechanics 2

Lecture notes

Guglielmo Mazzola



Master in Complex Systems
SISSA, Trieste, Italy

Abstract

Contents

1 Approximate solutions: variational and perturbation theory **1**

1.1 Setting and notation 1
1.2 Finite basis representation 2
1.3 The variational principle 3
1.3.1 Proof by expansion in the exact eigenbasis 3
1.3.2 Variational parameters 5
1.4 Example: continuum variational calculation 5
1.5 Time-independent perturbation theory 6
1.5.1 Small perturbations 6
1.5.2 Derivation for a non-degenerate level 7
1.6 Example: continuum perturbation calculation 9
1.6.1 The same oscillator with a variational wavefunction 10
1.7 Degenerate perturbation theory 11
1.7.1 A two-dimensional toy model 11
1.8 Derivation with the choice of a good basis 12
1.9 Example: one spin-1/2 system 16
1.9.1 Variational treatment 16
1.9.2 Non-degenerate perturbative treatment 17
1.9.3 Degenerate spin example 18
1.10 Comparison of the two methods 20

Chapter 1

Approximate solutions: variational and perturbation theory

This chapter presents two approximation methods in time-independent quantum mechanics: the variational principle and stationary perturbation theory. The variational principle is an optimization method based on trial states and gives a rigorous upper bound to the ground-state energy. Perturbation theory is an expansion method based on a solvable Hamiltonian plus a small correction.

1.1 Setting and notation

Let H be a self-adjoint Hamiltonian acting on a Hilbert space $\mathcal{H}$, with a lowest energy E_0. Its exact stationary states satisfy

$$H\left|\Psi_n\right\rangle = E_n\left|\Psi_n\right\rangle. \tag{1.1}$$

When the discrete spectrum can be ordered, we write

$$E_0 \leq E_1 \leq E_2 \leq \cdots. \tag{1.2}$$

Exact eigenstates of the full Hamiltonian are denoted by $|\Psi_n\rangle$. Trial states used in the variational method are denoted by $|\phi\rangle$ or $|\phi(\boldsymbol{a})\rangle$, where $\boldsymbol{a}$ is a set of variational parameters. In perturbation theory the Hamiltonian is written as

$$H(\lambda) = H_0 + \lambda V. \tag{1.3}$$

Here H_0 is exactly solvable, V is the perturbation, and λ is a dimensionless parameter. It may measure the strength of an applied field relative to a fixed scale, or it may simply be a bookkeeping parameter used to order the calculation.
The unperturbed eigenvalue problem is

$$H_0\left|n^{(0)}\right\rangle = E_n^{(0)}\left|n^{(0)}\right\rangle. \tag{1.4}$$

Unless otherwise stated, the unperturbed states are orthonormal:

$$\left\langle m^{(0)}\middle|n^{(0)}\right\rangle = \delta_{mn}. \tag{1.5}$$

Matrix elements of the perturbation in this basis are abbreviated as

$$V_{mn} = \left\langle m^{(0)} \middle| V \middle| n^{(0)} \right\rangle . \tag{1.6}$$

1.2 Finite basis representation

Both approximation methods are most transparent when a basis is fixed. Let $\{|\chi_i\rangle\}_{i=1}^{M}$ be a finite set of linearly independent vectors. A general state in the subspace spanned by these vectors is

$$|\phi\rangle = \sum_{i=1}^{M} c_i \, |\chi_i\rangle . \tag{1.7}$$

If the basis is not orthonormal, the overlap matrix is

$$S_{ij} = \langle \chi_i | \chi_j \rangle . \tag{1.8}$$

The Hamiltonian matrix in the same basis is

$$H_{ij} = \langle \chi_i | H | \chi_j \rangle . \tag{1.9}$$

The norm of the state is

$$\langle \phi | \phi \rangle = \sum_{i,j=1}^{M} c_i^* S_{ij} c_j . \tag{1.10}$$

Its expectation value of energy is

$$\mathcal{E}[\boldsymbol{c}] = \frac{\sum_{i,j=1}^{M} c_i^* H_{ij} c_j}{\sum_{i,j=1}^{M} c_i^* S_{ij} c_j} . \tag{1.11}$$

In matrix notation this is

$$\mathcal{E}[\boldsymbol{c}] = \frac{\boldsymbol{c}^\dagger \boldsymbol{H} \boldsymbol{c}}{\boldsymbol{c}^\dagger \boldsymbol{S} \boldsymbol{c}} . \tag{1.12}$$

Stationarity with respect to the independent coefficients c_i^* gives the generalized eigenvalue equation

$$\boldsymbol{H}\boldsymbol{c} = \mathcal{E}\boldsymbol{S}\boldsymbol{c} . \tag{1.13}$$

If the basis is orthonormal, $\boldsymbol{S} = \boldsymbol{I}$ and the equation reduces to

$$\boldsymbol{H}\boldsymbol{c} = \mathcal{E}\boldsymbol{c}, \tag{1.14}$$

This is the Schrödinger equation restricted to the chosen subspace. Its lowest eigenvalue gives our best ground-state energy within that subspace. Enlarging the basis while keeping the previous basis vectors can only improve this estimate or leave it unchanged.
The continuum position representation is obtained by taking the basis to be $|x\rangle$. A wave-function is

$$\phi(x) = \langle x | \phi \rangle . \tag{1.15}$$

For a one-dimensional Hamiltonian with local potential $U(x)$,

$$H = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + U(x), \tag{1.16}$$

the energy functional is

$$\mathcal{E}[\phi] = \frac{\int_{-\infty}^{\infty} \phi^*(x)\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + U(x)\right]\phi(x)\mathrm{d}x}{\int_{-\infty}^{\infty}|\phi(x)|^2\mathrm{d}x}. \tag{1.17}$$

The discrete and continuum formulas are the same statement in different bases: sums over labels become integrals over continuous variables.

1.3 The variational principle

The variational principle is a theorem about expectation values. It does not assume that the Hamiltonian is close to another Hamiltonian, and it does not involve a small parameter. It says that every normalized trial state gives an upper bound to the exact ground-state energy. Let $|\phi\rangle$ be a normalized trial state with finite energy and the required boundary conditions,

$$\langle\phi|\phi\rangle = 1. \tag{1.18}$$

Its variational energy is

$$\mathcal{E}[\phi] = \langle\phi|H|\phi\rangle\,. \tag{1.19}$$

The variational principle states

$$\mathcal{E}[\phi] \geq E_0. \tag{1.20}$$

1.3.1 Proof by expansion in the exact eigenbasis

The proof is especially simple when the trial state is expanded in the eigenstates of the same Hamiltonian H whose ground-state energy is sought. Assume first that the relevant spectrum is discrete and that the exact eigenstates form an orthonormal basis in the sector of Hilbert space under consideration:

$$H\,|\Psi_n\rangle = E_n\,|\Psi_n\rangle\,, \qquad \langle\Psi_m|\Psi_n\rangle = \delta_{mn}. \tag{1.21}$$

A generic normalized state can then be written as

$$|\phi\rangle = \sum_n a_n\,|\Psi_n\rangle\,. \tag{1.22}$$

In the position representation this same statement reads

$$\phi(x) = \sum_n a_n\Psi_n(x), \qquad \Psi_n(x) = \langle x|\Psi_n\rangle\,. \tag{1.23}$$

Orthonormality gives the coefficients as projections,

$$a_n = \langle\Psi_n|\phi\rangle\,, \tag{1.24}$$

and normalization gives

$$\sum_n |a_n|^2 = 1. \tag{1.25}$$

The expectation value of the Hamiltonian is

$$\mathcal{E}[\phi] = \langle\phi|H|\phi\rangle = \sum_{m,n} a_m^* a_n \langle\Psi_m|H|\Psi_n\rangle . \tag{1.26}$$

Since $H|\Psi_n\rangle = E_n|\Psi_n\rangle$, the matrix element is diagonal in the exact eigenbasis:

$$\langle\Psi_m|H|\Psi_n\rangle = E_n \langle\Psi_m|\Psi_n\rangle = E_n\delta_{mn}. \tag{1.27}$$

Therefore

$$\mathcal{E}[\phi] = \sum_{m,n} a_m^* a_n E_n \delta_{mn} = \sum_n |a_n|^2 E_n. \tag{1.28}$$

Subtracting the exact ground-state energy gives the most useful form of the result:

$$\mathcal{E}[\phi] - E_0 = \sum_n |a_n|^2 (E_n - E_0). \tag{1.29}$$

Every term on the right-hand side is non-negative because $E_n \geq E_0$ and $|a_n|^2 \geq 0$. Hence

$$\mathcal{E}[\phi] \geq E_0. \tag{1.30}$$

The proof also explains the physical meaning of the bound. The numbers $|a_n|^2$ are the weights of the exact energy eigenstates in the trial state, so $\mathcal{E}[\phi]$ is a weighted average of exact energies. A weighted average cannot be lower than the smallest energy appearing in the spectrum.

The same argument also covers unnormalized trial states, which are common in calculations. If

$$|\phi\rangle = \sum_n a_n |\Psi_n\rangle \tag{1.31}$$

but $\sum_n |a_n|^2$ is not assumed to be one, then the Rayleigh quotient is

$$\mathcal{E}[\phi] = \frac{\langle\phi|H|\phi\rangle}{\langle\phi|\phi\rangle} = \frac{\sum_n |a_n|^2 E_n}{\sum_n |a_n|^2}. \tag{1.32}$$

This can be rewritten as

$$\mathcal{E}[\phi] = E_0 + \frac{\sum_n |a_n|^2 (E_n - E_0)}{\sum_n |a_n|^2}. \tag{1.33}$$

Thus again

$$\mathcal{E}[\phi] \geq E_0. \tag{1.34}$$

Equality is obtained only when the trial state lies entirely in the ground-state eigenspace. If the ground state is non-degenerate, this means $|\phi\rangle$ must be proportional to $|\Psi_0\rangle$. If the ground state is degenerate, any superposition inside the exact ground-state subspace gives equality. Thus the method gives an upper bound, but the quality of the bound depends on how well the chosen family of trial states can approximate the true ground state.

1.3.2 Variational parameters

Suppose the trial state depends on real parameters $\boldsymbol{a} = (a_1, a_2, \ldots, a_p)$. The variational estimate is

$$E_{\mathrm{var}} = \min_{\boldsymbol{a}} \mathcal{E}[\phi(\boldsymbol{a})]. \tag{1.35}$$

At an interior minimum the stationarity conditions are

$$\frac{\partial \mathcal{E}[\phi(\boldsymbol{a})]}{\partial a_k} = 0, \tag{1.36}$$

for all k. The resulting number remains an upper bound:

$$E_{\mathrm{var}} \geq E_0. \tag{1.37}$$

In practice, the parameters often have a physical interpretation. A width parameter may represent the spatial size of a bound state, while a mixing angle may represent the orientation of a spinor. The minimization chooses the best such value within the restricted family.

1.4 Example: continuum variational calculation

Consider the one-dimensional harmonic oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2. \tag{1.38}$$

Choose a normalized Gaussian trial wavefunction,

$$\phi_a(x) = \left(\frac{2a}{\pi}\right)^{1/4} \exp\bigl(-a x^2\bigr), \tag{1.39}$$

with $a > 0$. The parameter a controls the width. Large a localizes the particle near the origin, increasing kinetic energy; small a spreads the particle out, increasing potential energy.
The two expectation values needed are

$$\bigl\langle x^2 \bigr\rangle = \frac{1}{4a}, \tag{1.40}$$

and

$$\bigl\langle p^2 \bigr\rangle = \hbar^2 a. \tag{1.41}$$

Therefore

$$\mathcal{E}(a) = \frac{\hbar^2 a}{2m} + \frac{m\omega^2}{8a}. \tag{1.42}$$

The stationary condition is

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}a} = \frac{\hbar^2}{2m} - \frac{m\omega^2}{8a^2} = 0. \tag{1.43}$$

Hence

$$a_\star = \frac{m\omega}{2\hbar}. \tag{1.44}$$

Substitution gives

$$E_{\text{var}} = \frac{\hbar\omega}{2}. \tag{1.45}$$

In this special case the trial family contains the exact ground state, so the upper bound is saturated. In a less ideal trial family the same procedure would still produce an upper bound, but it would not generally be exact.

Connection with Machine Learning: For more complicated systems, the trial wavefunction may have thousands of parameters, and its energy is evaluated numerically. A neural network is one possible way to represent such a wavefunction. What plays the role of the model, the trainable parameters, and the loss function in this calculation? Notice that we can minimize the energy without being given examples of the exact ground-state wavefunction.

1.5 Time-independent perturbation theory

Perturbation theory addresses a different problem. Instead of optimizing over trial states, we assume that the Hamiltonian is close to one that we can solve exactly. The central object is

$$H(\lambda) = H_0 + \lambda V. \tag{1.46}$$

The exact n-th eigenvalue and eigenstate connected continuously to the unperturbed state $\left|n^{(0)}\right\rangle$ are assumed to admit expansions in powers of λ:

$$E_n(\lambda) = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots, \tag{1.47}$$

and

$$|n(\lambda)\rangle = \left|n^{(0)}\right\rangle + \lambda \left|n^{(1)}\right\rangle + \lambda^2 \left|n^{(2)}\right\rangle + \cdots. \tag{1.48}$$

The coefficients $E_n^{(k)}$ and $\left|n^{(k)}\right\rangle$ do not depend on λ. We determine them by inserting these expansions into the Schrödinger equation and comparing equal powers of λ.
The series need not converge for this procedure to be useful. Sometimes the first few terms give an accurate answer at small λ, even though adding more and more terms eventually makes it worse. Such a series is called asymptotic. The quartic oscillator below is an example. For now, we consider a non-degenerate level; degenerate levels will be treated separately.

1.5.1 Small perturbations

Saying that V is small is not simply a statement about the numerical size of the matrix entries of V. In quantum mechanics, the effect of an off-diagonal matrix element must be compared with the energy spacing of the states it connects. In the unperturbed eigenbasis,

$$H_{mn}(\lambda) = E_n^{(0)}\delta_{mn} + \lambda V_{mn}. \tag{1.49}$$

For a non-degenerate level n, the dimensionless mixing parameter associated with another level m is

$$\epsilon_{mn} = \left|\frac{\lambda V_{mn}}{E_n^{(0)} - E_m^{(0)}}\right|. \tag{1.50}$$

As a first check, we require

$$\epsilon_{mn} \ll 1 \tag{1.51}$$

for the states that are appreciably coupled by V. A small matrix element can still produce strong mixing if the energy gap is even smaller. When many states contribute, we should also check that their combined correction to the wavefunction is small.
The same point can be seen in a two-level subspace spanned by $\left|n^{(0)}\right\rangle$ and $\left|m^{(0)}\right\rangle$. To keep the example simple, take $V_{nn} = V_{mm} = 0$. After subtracting an irrelevant average energy, the Hamiltonian in this subspace has the form

$$\begin{pmatrix} \Delta/2 & \lambda V_{nm} \\ \lambda V_{mn} & -\Delta/2 \end{pmatrix}, \tag{1.52}$$

where

$$\Delta = E_n^{(0)} - E_m^{(0)}. \tag{1.53}$$

If $|\lambda V_{mn}| \ll |\Delta|$, the eigenvectors are only weakly rotated from the unperturbed basis. If $\Delta = 0$, however, the ratio is not small for any nonzero coupling. This is the elementary reason why degeneracy requires a different treatment.

1.5.2 Derivation for a non-degenerate level

First of all, let's adopt the following *intermediate normalization* convention. For a non-degenerate unperturbed level, a convenient convention [1] is intermediate normalization:

$$\left\langle n^{(0)} \middle| n(\lambda) \right\rangle = 1. \tag{1.54}$$

Inserting the expansion of the state gives

$$\left\langle n^{(0)} \middle| n^{(k)} \right\rangle = 0 \tag{1.55}$$

for every $k \geq 1$. The correction $\left|n^{(1)}\right\rangle$ therefore has no component along $\left|n^{(0)}\right\rangle$. This convention fixes the otherwise arbitrary normalization of the perturbed eigenvector order by order.
Insert the expansions into the eigenvalue equation

$$(H_0 + \lambda V) \left|n(\lambda)\right\rangle = E_n(\lambda) \left|n(\lambda)\right\rangle. \tag{1.56}$$

The coefficient of λ^0 gives the unperturbed equation

$$H_0 \left|n^{(0)}\right\rangle = E_n^{(0)} \left|n^{(0)}\right\rangle. \tag{1.57}$$

[1] Eigenstates are defined only up to an overall multiplicative factor: if $|n(\lambda)\rangle$ is an eigenstate, then so is $c(\lambda)\,|n(\lambda)\rangle$ for any nonzero function $c(\lambda)$. Intermediate normalization removes this ambiguity by fixing the overlap with the unperturbed state to be unity, $\left\langle n^{(0)} \middle| n(\lambda) \right\rangle = 1$. Consequently, the higher-order corrections satisfy $\left\langle n^{(0)} \middle| n^{(k)} \right\rangle = 0$ for all $k \geq 1$. Note that this condition is distinct from the usual normalization $\langle n(\lambda)|n(\lambda)\rangle = 1$; under intermediate normalization, the exact state generally has norm $1 + \mathcal{O}(\lambda^2)$ and can be rescaled to unit norm after the perturbative corrections have been determined.

The coefficient of λ^1 gives

$$(H_0 - E_n^{(0)}) \left|n^{(1)}\right\rangle = (E_n^{(1)} - V) \left|n^{(0)}\right\rangle. \tag{1.58}$$

To extract the energy shift, multiply by $\left\langle n^{(0)}\right|$ on the left. The left-hand side vanishes because $\left\langle n^{(0)}\right| (H_0 - E_n^{(0)}) = 0$, so

$$0 = E_n^{(1)} - V_{nn}. \tag{1.59}$$

Therefore

$$E_n^{(1)} = V_{nn} = \left\langle n^{(0)}\middle|V\middle|n^{(0)}\right\rangle. \tag{1.60}$$

To find the first correction to the state, multiply the first-order equation by $\left\langle m^{(0)}\right|$ with $m \neq n$:

$$(E_m^{(0)} - E_n^{(0)}) \left\langle m^{(0)}\middle|n^{(1)}\right\rangle = -V_{mn}. \tag{1.61}$$

Thus

$$\left\langle m^{(0)}\middle|n^{(1)}\right\rangle = \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}}. \tag{1.62}$$

Because intermediate normalization removes the component along $\left|n^{(0)}\right\rangle$, the full first-order correction is

$$\left|n^{(1)}\right\rangle = \sum_{m\neq n} \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} \left|m^{(0)}\right\rangle. \tag{1.63}$$

The coefficient of λ^2 in the Schrodinger equation is

$$(H_0 - E_n^{(0)}) \left|n^{(2)}\right\rangle = (E_n^{(1)} - V) \left|n^{(1)}\right\rangle + E_n^{(2)} \left|n^{(0)}\right\rangle. \tag{1.64}$$

Multiplying by $\left\langle n^{(0)}\right|$ gives

$$0 = E_n^{(1)} \left\langle n^{(0)}\middle|n^{(1)}\right\rangle - \left\langle n^{(0)}\middle|V\middle|n^{(1)}\right\rangle + E_n^{(2)}. \tag{1.65}$$

The first term vanishes by intermediate normalization, so

$$E_n^{(2)} = \left\langle n^{(0)}\middle|V\middle|n^{(1)}\right\rangle. \tag{1.66}$$

Substituting the expression for $\left|n^{(1)}\right\rangle$ gives

$$E_n^{(2)} = \sum_{m\neq n} \frac{V_{nm}V_{mn}}{E_n^{(0)} - E_m^{(0)}}. \tag{1.67}$$

Since V is Hermitian, $V_{nm}V_{mn} = |V_{mn}|^2$, and therefore

$$E_n^{(2)} = \sum_{m\neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}. \tag{1.68}$$

The perturbed energy through second order is

$$E_n(\lambda) = E_n^{(0)} + \lambda V_{nn} + \lambda^2 \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}} + O(\lambda^3). \tag{1.69}$$

For the ground state, every denominator in the second-order term is negative, so

$$E_0^{(2)} \leq 0. \tag{1.70}$$

This sign has a simple interpretation: mixing the unperturbed ground state with excited states can lower the energy of the approximate state for the full Hamiltonian. It is also consistent with the variational principle, although the two methods are conceptually distinct.

1.6 Example: continuum perturbation calculation

Consider the harmonic oscillator perturbed by a weak quartic potential,

$$H(\lambda) = H_0 + \lambda V, \tag{1.71}$$

with

$$H_0 = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2, \tag{1.72}$$

and

$$V = \hbar\omega \left(\frac{x}{\ell}\right)^4, \qquad \ell = \sqrt{\frac{\hbar}{m\omega}}. \tag{1.73}$$

The factor $\hbar\omega/\ell^4$ keeps λ dimensionless. We take $\lambda \geq 0$, so the potential remains confining. The quartic term makes the potential steeper away from the origin. The unperturbed ground-state energy is

$$E_0^{(0)} = \frac{1}{2}\hbar\omega. \tag{1.74}$$

The first-order correction is

$$E_0^{(1)} = \left\langle 0^{(0)} \middle| V \middle| 0^{(0)} \right\rangle = \frac{\hbar\omega}{\ell^4} \left\langle x^4 \right\rangle_0 . \tag{1.75}$$

The harmonic-oscillator ground-state probability density is Gaussian and satisfies

$$\left\langle x^2 \right\rangle_0 = \frac{\ell^2}{2}. \tag{1.76}$$

For a Gaussian distribution,

$$\left\langle x^4 \right\rangle_0 = 3 \left\langle x^2 \right\rangle_0^2 . \tag{1.77}$$

Therefore

$$E_0^{(1)} = \frac{3}{4}\hbar\omega. \tag{1.78}$$

The ground-state energy through first order is

$$E_0(\lambda) = \hbar\omega \left[\frac{1}{2} + \frac{3}{4}\lambda + O(\lambda^2)\right]. \tag{1.79}$$

The first-order shift is just an expectation value in the unperturbed state. To go one order further, we need the states coupled to the ground state by x^4. Applying the oscillator raising and lowering operators gives

$$\frac{x^4}{\ell^4}\left|0^{(0)}\right\rangle = \frac{3}{4}\left|0^{(0)}\right\rangle + \frac{3\sqrt{2}}{2}\left|2^{(0)}\right\rangle + \frac{\sqrt{6}}{2}\left|4^{(0)}\right\rangle. \tag{1.80}$$

Only the levels $n = 2$ and $n = 4$ enter the second-order energy correction:

$$E_0^{(2)} = \frac{(\hbar\omega)^2(9/2)}{-2\hbar\omega} + \frac{(\hbar\omega)^2(3/2)}{-4\hbar\omega} = -\frac{21}{8}\hbar\omega. \tag{1.81}$$

We therefore obtain

$$E_0(\lambda) = \hbar\omega\left[\frac{1}{2} + \frac{3}{4}\lambda - \frac{21}{8}\lambda^2 + O(\lambda^3)\right]. \tag{1.82}$$

1.6.1 The same oscillator with a variational wavefunction

Let us now solve the same problem using the Gaussian trial state from the previous example:

$$\phi_a(x) = \left(\frac{2a}{\pi}\right)^{1/4} e^{-ax^2}, \qquad a > 0. \tag{1.83}$$

The kinetic and harmonic terms have already been calculated. For the quartic term we only need

$$\left\langle x^4\right\rangle_a = 3\left\langle x^2\right\rangle_a^2 = \frac{3}{16a^2}. \tag{1.84}$$

The variational energy is therefore

$$\mathcal{E}(a) = \frac{\hbar^2 a}{2m} + \frac{m\omega^2}{8a} + \frac{3\lambda\hbar\omega}{16a^2\ell^4}. \tag{1.85}$$

We find the optimal width by setting its derivative to zero:

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}a} = \frac{\hbar^2}{2m} - \frac{m\omega^2}{8a^2} - \frac{3\lambda\hbar\omega}{8a^3\ell^4} = 0. \tag{1.86}$$

At $\lambda = 0$ we recover $a_0 = m\omega/(2\hbar)$. For $\lambda > 0$, the optimal a is larger: the wavefunction becomes narrower because the quartic term penalizes large values of $|x|$. The minimum can be found numerically for any positive λ.

There is already a useful connection with perturbation theory. If we keep the original harmonic-oscillator width a_0, the variational energy is exactly the first-order perturbative estimate. Allowing the width to change can only lower this energy, so

$$E_0(\lambda) \le E_{\mathrm{var}}(\lambda) \le \hbar\omega\left(\frac{1}{2} + \frac{3}{4}\lambda\right). \tag{1.87}$$

For a closer comparison at small λ, solving the stationarity equation to first order gives

$$a_\star = \frac{m\omega}{2\hbar}\left[1 + 3\lambda + O(\lambda^2)\right]. \tag{1.88}$$

Substituting this result into $\mathcal{E}(a)$ gives

$$E_{\text{var}}(\lambda) = \hbar\omega \left[\frac{1}{2} + \frac{3}{4}\lambda - \frac{9}{4}\lambda^2 + O(\lambda^3)\right]. \tag{1.89}$$

The two methods agree at first order. At second order, changing the Gaussian width captures part of the energy lowering, but the coefficient differs from $-21/8$: the true wavefunction also changes its shape. Unlike the harmonic case, a Gaussian is no longer exact.
The full minimized function $\mathcal{E}(a_\star)$ remains an upper bound even when λ is not small. Truncating a perturbation series does not, in general, give an upper bound. The first-order estimate above is an upper bound because it is itself the expectation value of H in the unperturbed ground state.

1.7 Degenerate perturbation theory

Degenerate perturbation theory is not a separate physical approximation; it is the correct perturbative expansion when the unperturbed Hamiltonian has an eigenspace of dimension greater than one. It is forced on us by the meaning of "small." In the non-degenerate formulas, smallness requires $|\lambda V_{mn}| \ll |E_n^{(0)} - E_m^{(0)}|$. Inside a degenerate subspace the denominator is zero, so no nonzero off-diagonal perturbation can be treated as a small mixing in an arbitrary basis.

1.7.1 A two-dimensional toy model

The basic issue is visible in a two-level model. Let

$$H_0 = E_d \mathbb{I}, \tag{1.90}$$

and choose

$$V = v\sigma_x, \qquad v \in \mathbb{R} \tag{1.91}$$

In the basis $\{|1\rangle, |2\rangle\}$, the full Hamiltonian is

$$\boldsymbol{H}(\lambda) = \begin{pmatrix} E_d & \lambda v \\ \lambda v & E_d \end{pmatrix}. \tag{1.92}$$

If one incorrectly applies the non-degenerate first-order energy formula to the basis vectors $|1\rangle$ and $|2\rangle$, one finds

$$\langle 1|V|1\rangle = \langle 2|V|2\rangle = 0. \tag{1.93}$$

That would suggest no first-order splitting. But the exact eigenvectors are

$$|+\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle), \tag{1.94}$$

and

$$|-\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle), \tag{1.95}$$

with exact energies

$$E_+(\lambda) = E_d + \lambda v, \tag{1.96}$$

and

$$E_-(\lambda) = E_d - \lambda v. \tag{1.97}$$

The apparent discontinuity of the eigenvectors is not a real discontinuity. At $\lambda = 0$ every linear combination of $|1\rangle$ and $|2\rangle$ is an equally valid eigenvector of H_0. The perturbation selects the particular combinations that diagonalize V within the degenerate subspace. Within an exactly degenerate subspace, a nonzero off-diagonal matrix element generally produces an order-one rotation of the eigenvectors, even though the resulting energy splitting is only of order λ.

Notice that, if the original hamiltonian is non-degenerate, then the energy correction would have been correcly at second order.

1.8 Derivation with the choice of a good basis

Consider again

$$H(\lambda) = H_0 + \lambda V, \tag{1.98}$$

where V is Hermitian. Suppose the unperturbed energy $E_n^{(0)}$ is N-fold degenerate, and denote its eigenspace by $\mathcal{V}_N$. The spectrum looks like: $E_0^{(0)} < E_1^{(0)} < \cdots < E_n^{(0)} = E_{n+1}^{(0)} = \cdots = E_{n+N-1}^{(0)} < E_{n+N}^{(0)} < \cdots$. We seek N orthonormal starting states

$$H_0 \left| n^{(0)}, k \right\rangle = E_n^{(0)} \left| n^{(0)}, k \right\rangle, \qquad \langle n^{(0)}, l | n^{(0)}, k \rangle = \delta_{lk}, \qquad k, l = 1, \ldots, N. \tag{1.99}$$

Their choice within $\mathcal{V}_N$ is not yet fixed: the perturbation will tell us which basis to use. Basis can be transformed using a linear operations. We denote the unperturbed states outside $\mathcal{V}_N$ by $\left| p^{(0)} \right\rangle$, with energies $E_p^{(0)} \neq E_n^{(0)}$. Thus

$$\mathcal{H} = \mathcal{V}_N \oplus \mathcal{V}_N^{\perp}. \tag{1.100}$$

For simplicity, we consider a discrete spectrum and an isolated unperturbed level $E_n^{(0)}$.

For each branch k, write

$$|n, k\rangle_\lambda = \left| n^{(0)}, k \right\rangle + \lambda \left| n^{(1)}, k \right\rangle + \lambda^2 \left| n^{(2)}, k \right\rangle + \cdots, \tag{1.101}$$

$$E_{n,k}(\lambda) = E_n^{(0)} + \lambda E_{n,k}^{(1)} + \lambda^2 E_{n,k}^{(2)} + \cdots. \tag{1.102}$$

We use intermediate normalization,

$$\langle n^{(0)}, k | n, k \rangle_\lambda = 1, \tag{1.103}$$

so that

$$\langle n^{(0)}, k | n^{(r)}, k \rangle = 0, \qquad r \geq 1. \tag{1.104}$$

This fixes the overlap with the starting state; it does not make the full perturbed state have unit norm. Most importantly, it removes only the correction along $\left| n^{(0)}, k \right\rangle$. The correction

can still have components along $\left|n^{(0)}, l\right\rangle$ with $l \neq k$, inside the same degenerate subspace, $\mathcal{V}_N$.

Substituting into the Schrödinger equation and matching powers of λ gives

$$(H_0 - E_n^{(0)}) \left|n^{(0)}, k\right\rangle = 0, \tag{1.105}$$

$$(H_0 - E_n^{(0)}) \left|n^{(1)}, k\right\rangle = (E_{n,k}^{(1)} - V) \left|n^{(0)}, k\right\rangle, \tag{1.106}$$

$$(H_0 - E_n^{(0)}) \left|n^{(2)}, k\right\rangle = (E_{n,k}^{(1)} - V) \left|n^{(1)}, k\right\rangle + E_{n,k}^{(2)} \left|n^{(0)}, k\right\rangle. \tag{1.107}$$

So far, the equations are as before. However, now, we will need both the first-order and the second-order equations to determine the complete first-order correction to the state. The $0 - th$ order equation is trivially satisfied and useless to us, so we start analyzing Eq 1.106

Step 1: the energy correction and the good basis. Multiply the first-order equation by $\left\langle n^{(0)}, l\right|$. The left-hand side vanishes because we are in the degenerate subspace:

$$\left\langle n^{(0)}, l\right| (H_0 - E_n^{(0)}) = 0. \tag{1.108}$$

Using orthonormality on the right-hand side, we obtain

$$0 = E_{n,k}^{(1)} \delta_{lk} - \left\langle n^{(0)}, l\right| V \left|n^{(0)}, k\right\rangle, \tag{1.109}$$

or

$$\boxed{\left\langle n^{(0)}, l\right| V \left|n^{(0)}, k\right\rangle = E_{n,k}^{(1)} \delta_{lk}.} \tag{1.110}$$

This equation tells us two things. For $l = k$, it gives the first-order energy correction:

$$E_{n,k}^{(1)} = \left\langle n^{(0)}, k\right| V \left|n^{(0)}, k\right\rangle. \tag{1.111}$$

For $l \neq k$, it requires

$$\left\langle n^{(0)}, l\right| V \left|n^{(0)}, k\right\rangle = 0. \tag{1.112}$$

Thus we cannot evaluate the diagonal matrix elements in just any unperturbed basis and stop there. **We must choose an orthonormal basis in $\mathcal{V}_N$ that makes the perturbation diagonal within that subspace.** Its eigenvalues give the first-order energy shifts, and its eigenvectors give the correct starting states $\left|n^{(0)}, k\right\rangle$.

Only the restriction of V to $\mathcal{V}_N$ is diagonalized. The perturbation can still connect these starting states to states outside $\mathcal{V}_N$. Through first order,

$$E_{n,k}(\lambda) = E_n^{(0)} + \lambda E_{n,k}^{(1)} + O(\lambda^2). \tag{1.113}$$

Operationally, if we had not chosen a basis that is diagonal, we must do so to proceed. Diagonalizing the perturbation automatically gives the energy shift that we need.

Finding the good basis from a symmetry. Sometimes we know the good basis in advance. Suppose a Hermitian operator T preserves $\mathcal{V}_N$ and commutes with V. For example, preservation of $\mathcal{V}_N$ follows if $[T, H_0] = 0$. If the eigenvalues of T within $\mathcal{V}_N$ are *distinct*, choose

$$T \left| n^{(0)}, k \right\rangle = t_k \left| n^{(0)}, k \right\rangle . \tag{1.114}$$

Then

$$0 = \left\langle n^{(0)}, l \right| [T, V] \left| n^{(0)}, k \right\rangle = (t_l - t_k) \left\langle n^{(0)}, l \right| V \left| n^{(0)}, k \right\rangle . \tag{1.115}$$

Since $t_l \neq t_k$ for $l \neq k$, the off-diagonal matrix elements vanish. The eigenbasis of T is therefore already a good basis. If T has repeated eigenvalues within $\mathcal{V}_N$, it need not fix the whole basis.

Step 2: the state correction outside the degenerate subspace. Finding the correction to the state is more difficult. We start by calculating the projection of the correction that lies outside the initial degenerate subspace. Now multiply the first-order equation by $\left\langle p^{(0)} \right|$, where $\left| p^{(0)} \right\rangle$ lies outside $\mathcal{V}_N$:

$$(E_p^{(0)} - E_n^{(0)}) \langle p^{(0)} | n^{(1)}, k \rangle = - \left\langle p^{(0)} \right| V \left| n^{(0)}, k \right\rangle . \tag{1.116}$$

The term containing $E_{n,k}^{(1)}$ disappears because $\langle p^{(0)} | n^{(0)}, k \rangle = 0$. Therefore

$$\langle p^{(0)} | n^{(1)}, k \rangle = \frac{\left\langle p^{(0)} \right| V \left| n^{(0)}, k \right\rangle}{E_n^{(0)} - E_p^{(0)}}. \tag{1.117}$$

The part of the state correction outside the degenerate subspace is

$$\boxed{\left| n^{(1)}, k \right\rangle_{\text{out}} = \sum_{p \notin \mathcal{V}_N} \frac{\left\langle p^{(0)} \right| V \left| n^{(0)}, k \right\rangle}{E_n^{(0)} - E_p^{(0)}} \left| p^{(0)} \right\rangle .} \tag{1.118}$$

Here $p \notin \mathcal{V}_N$ means that we exclude *all* states at the unperturbed energy $E_n^{(0)}$. This has the same form as the non-degenerate result, but it is not necessarily the complete first-order correction.

Step 3: the state correction inside the degenerate subspace. Write the full correction as

$$\left| n^{(1)}, k \right\rangle = \left| n^{(1)}, k \right\rangle_{\text{out}} + \left| n^{(1)}, k \right\rangle_{\text{in}} . \tag{1.119}$$

The first-order equation cannot determine the second term: $H_0 - E_n^{(0)}$ annihilates every vector in $\mathcal{V}_N$. We must therefore resort to the second-order equation, Eq. (1.107), and multiply it by $\left\langle n^{(0)}, l \right|$. Again the left-hand side vanishes:

$$0 = E_{n,k}^{(1)} \langle n^{(0)}, l | n^{(1)}, k \rangle - \left\langle n^{(0)}, l \right| V \left| n^{(1)}, k \right\rangle + E_{n,k}^{(2)} \delta_{lk}. \tag{1.120}$$

To evaluate the middle term, separate the inside and outside parts. For the inside part, insert the basis of $\mathcal{V}_N$ and use Eq. (1.110):

$$\left\langle n^{(0)}, l\right| V \left|n^{(1)}, k\right\rangle_{\text{in}} = \sum_{q=1}^{N} \left\langle n^{(0)}, l\right| V \left|n^{(0)}, q\right\rangle \langle n^{(0)}, q|n^{(1)}, k\rangle \tag{1.121}$$

$$= \sum_{q=1}^{N} E^{(1)}_{n,q}\delta_{lq}\langle n^{(0)}, q|n^{(1)}, k\rangle \tag{1.122}$$

$$= E^{(1)}_{n,l}\langle n^{(0)}, l|n^{(1)}, k\rangle. \tag{1.123}$$

The projected second-order equation consequently becomes

$$0 = \left(E^{(1)}_{n,k} - E^{(1)}_{n,l}\right)\langle n^{(0)}, l|n^{(1)}, k\rangle - \left\langle n^{(0)}, l\right| V \left|n^{(1)}, k\right\rangle_{\text{out}} + E^{(2)}_{n,k}\delta_{lk}. \tag{1.124}$$

Taking $l = k$: the second-order energy. The first term vanishes, giving

$$E^{(2)}_{n,k} = \left\langle n^{(0)}, k\right| V \left|n^{(1)}, k\right\rangle_{\text{out}} \tag{1.125}$$

$$= \sum_{p\notin\mathcal{V}_N} \frac{\left\langle n^{(0)}, k\right| V \left|p^{(0)}\right\rangle \left\langle p^{(0)}\right| V \left|n^{(0)}, k\right\rangle}{E^{(0)}_n - E^{(0)}_p} \tag{1.126}$$

$$= \boxed{\sum_{p\notin\mathcal{V}_N} \frac{|\left\langle p^{(0)}\right| V \left|n^{(0)}, k\right\rangle|^2}{E^{(0)}_n - E^{(0)}_p}}. \tag{1.127}$$

In the last step we used Hermiticity of V; the matrix elements need not be real.

Taking $l \neq k$: the missing state correction. Suppose first that the degeneracy is completely lifted at first order, so $E^{(1)}_{n,k} \neq E^{(1)}_{n,l}$ for $k \neq l$. Equation (1.124) then gives

$$\boxed{\langle n^{(0)}, l|n^{(1)}, k\rangle = \frac{\left\langle n^{(0)}, l\right| V \left|n^{(1)}, k\right\rangle_{\text{out}}}{E^{(1)}_{n,k} - E^{(1)}_{n,l}}.} \tag{1.128}$$

Substituting the result of step 2,

$$\langle n^{(0)}, l|n^{(1)}, k\rangle = \sum_{p\notin\mathcal{V}_N} \frac{\left\langle n^{(0)}, l\right| V \left|p^{(0)}\right\rangle \left\langle p^{(0)}\right| V \left|n^{(0)}, k\right\rangle}{\left(E^{(1)}_{n,k} - E^{(1)}_{n,l}\right)\left(E^{(0)}_n - E^{(0)}_p\right)}. \tag{1.129}$$

Thus

$$\boxed{\left|n^{(1)}, k\right\rangle_{\text{in}} = \sum_{\substack{l=1\\ l\neq k}}^{N} \frac{\left\langle n^{(0)}, l\right| V \left|n^{(1)}, k\right\rangle_{\text{out}}}{E^{(1)}_{n,k} - E^{(1)}_{n,l}} \left|n^{(0)}, l\right\rangle.} \tag{1.130}$$

There is no component along $\left|n^{(0)}, k\right\rangle$ because of intermediate normalization.

The first-order corrected state is therefore

$$|n,k\rangle_\lambda = \left|n^{(0)},k\right\rangle + \lambda\left(\left|n^{(1)},k\right\rangle_{\text{out}} + \left|n^{(1)},k\right\rangle_{\text{in}}\right) + O(\lambda^2). \tag{1.131}$$

The inside contribution describes mixing through intermediate states outside the degenerate subspace. Although it is found from the second-order equation, it multiplies λ in the state expansion and is a *first-order state correction*.

If the degeneracy is not fully lifted. Step 1 still gives the correct first-order energy shifts, but Eq. (1.130) cannot be used between states with equal $E^{(1)}_{n,k}$. Within each remaining degenerate block, diagonalize the second-order matrix

$$M_{lk} = \sum_{p\notin\mathcal{V}_N} \frac{\left\langle n^{(0)},l\right| V \left|p^{(0)}\right\rangle \left\langle p^{(0)}\right| V \left|n^{(0)},k\right\rangle}{E^{(0)}_n - E^{(0)}_p}. \tag{1.132}$$

Its eigenvectors select the starting states within that block and its eigenvalues give their second-order energy shifts. One continues to higher order if needed; the same rule is applied at the first order that distinguishes the states. Determining state mixing within a block split only at second order generally requires the third-order equation.

1.9 Example: one spin-1/2 system

Let a spin-1/2 particle be described by the basis of σ_z eigenstates:

$$\sigma_z\left|\uparrow\right\rangle = \left|\uparrow\right\rangle, \tag{1.133}$$

and

$$\sigma_z\left|\downarrow\right\rangle = -\left|\downarrow\right\rangle. \tag{1.134}$$

Consider first the Hamiltonian

$$H = -\frac{\Delta}{2}\sigma_z + \gamma\sigma_x, \tag{1.135}$$

with $\Delta > 0$ and real γ. We will treat $\gamma\sigma_x$ as the perturbation. In the basis $\{\left|\uparrow\right\rangle, \left|\downarrow\right\rangle\}$, the matrix is

$$\boldsymbol{H} = \begin{pmatrix} -\Delta/2 & \gamma \\ \gamma & \Delta/2 \end{pmatrix}. \tag{1.136}$$

1.9.1 Variational treatment

Apart from an overall phase, a general normalized spinor can be written as

$$|\phi(\theta,\varphi)\rangle = \cos\frac{\theta}{2}\left|\uparrow\right\rangle + e^{\mathrm{i}\varphi}\sin\frac{\theta}{2}\left|\downarrow\right\rangle. \tag{1.137}$$

In this state,

$$\langle\sigma_z\rangle = \cos\theta, \tag{1.138}$$

and

$$\langle\sigma_x\rangle = \sin\theta\cos\varphi. \tag{1.139}$$

Therefore

$$\mathcal{E}(\theta,\varphi) = -\frac{\Delta}{2}\cos\theta + \gamma\sin\theta\cos\varphi. \tag{1.140}$$

For $\gamma > 0$, the minimum occurs when

$$\cos\varphi = -1. \tag{1.141}$$

The remaining energy function is

$$\mathcal{E}(\theta) = -\frac{\Delta}{2}\cos\theta - \gamma\sin\theta. \tag{1.142}$$

The optimal angle lies between 0 and $\pi/2$ and satisfies

$$\tan\theta_\star = \frac{2\gamma}{\Delta}. \tag{1.143}$$

The minimum energy is

$$E_{\text{var}} = -\sqrt{\frac{\Delta^2}{4} + \gamma^2}. \tag{1.144}$$

Since the trial state was the most general normalized spinor, the variational answer is the exact ground-state energy.

1.9.2 Non-degenerate perturbative treatment

We temporarily introduce the bookkeeping parameter λ and write $H(\lambda) = H_0 + \lambda V$. The physical Hamiltonian is recovered at $\lambda = 1$. Here

$$H_0 = -\frac{\Delta}{2}\sigma_z, \tag{1.145}$$

and

$$V = \gamma\sigma_x. \tag{1.146}$$

The unperturbed ground state is $|\uparrow\rangle$ with energy

$$E_\uparrow^{(0)} = -\frac{\Delta}{2}. \tag{1.147}$$

The excited state is $|\downarrow\rangle$ with energy

$$E_\downarrow^{(0)} = \frac{\Delta}{2}. \tag{1.148}$$

The first-order correction to the ground-state energy is

$$E_\uparrow^{(1)} = \langle\uparrow|\gamma\sigma_x|\uparrow\rangle = 0. \tag{1.149}$$

The second-order correction is

$$E_\uparrow^{(2)} = \frac{|\,\langle\downarrow|\gamma\sigma_x|\uparrow\rangle\,|^2}{E_\uparrow^{(0)} - E_\downarrow^{(0)}}. \tag{1.150}$$

Since

$$\langle\downarrow|\sigma_x|\uparrow\rangle = 1, \tag{1.151}$$

this becomes

$$E_{\uparrow}^{(2)} = -\frac{\gamma^2}{\Delta}. \tag{1.152}$$

Thus

$$E_{\uparrow}(\lambda) = -\frac{\Delta}{2} - \frac{\lambda^2\gamma^2}{\Delta} + O(\lambda^4). \tag{1.153}$$

Expanding the exact result confirms the perturbative answer:

$$-\sqrt{\frac{\Delta^2}{4} + \lambda^2\gamma^2} = -\frac{\Delta}{2} - \frac{\lambda^2\gamma^2}{\Delta} + \frac{\lambda^4\gamma^4}{\Delta^3} + O(\lambda^6). \tag{1.154}$$

Notice that the second-order estimate lies below the exact energy. This illustrates why a truncated perturbative energy is not a variational upper bound. The first-order correction to the ground-state vector is

$$\left|\uparrow^{(1)}\right\rangle = \frac{\langle\downarrow|V|\uparrow\rangle}{E_{\uparrow}^{(0)} - E_{\downarrow}^{(0)}} \left|\downarrow\right\rangle. \tag{1.155}$$

Therefore

$$|\uparrow(\lambda)\rangle = |\uparrow\rangle - \lambda\frac{\gamma}{\Delta}|\downarrow\rangle + O(\lambda^2). \tag{1.156}$$

The perturbative smallness condition is

$$\left|\frac{\lambda\gamma}{\Delta}\right| \ll 1. \tag{1.157}$$

Setting $\lambda = 1$ gives the condition $|\gamma| \ll \Delta$ for the original Hamiltonian.

1.9.3 Degenerate spin example

Let us return to the initial example that motivated treating the degenerate pertubation theory with extra care. We use the same notation of Sect. 1.8, so we retain the subscript n, even if the degenerate subspace is the whole Hilbert space (so $n = 0$). We do so to help familiarize with the notation.

$$H_0 = E_n^{(0)} I, \qquad V = v\sigma_x, \quad v \in \mathbb{R}, \quad v \neq 0. \tag{1.158}$$

The unperturbed level is two-fold degenerate: $N = 2$. In the usual S_z basis $\{|\uparrow\rangle, |\downarrow\rangle\}$,

$$V = \begin{pmatrix} 0 & v \\ v & 0 \end{pmatrix}, \qquad H(\lambda) = \begin{pmatrix} E_n^{(0)} & \lambda v \\ \lambda v & E_n^{(0)} \end{pmatrix}. \tag{1.159}$$

Both diagonal matrix elements of V vanish in this basis. However, its off-diagonal elements do not vanish, so this is not the good basis required by step 1.

Step 1: diagonalize V in V_2. The first-order shifts satisfy

$$\det\Big(V - E^{(1)} I\Big) = \begin{vmatrix} -E^{(1)} & v \\ v & -E^{(1)} \end{vmatrix} = (E^{(1)})^2 - v^2 = 0. \tag{1.160}$$

Hence

$$E^{(1)}_{n,1} = v, \qquad E^{(1)}_{n,2} = -v. \tag{1.161}$$

For the eigenvalue v, the equation

$$\begin{pmatrix} 0 & v \\ v & 0 \end{pmatrix} \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} = v \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix} \tag{1.162}$$

requires $c_\downarrow = c_\uparrow$. For the eigenvalue $-v$, it requires $c_\downarrow = -c_\uparrow$. After normalization, the good starting states are

$$\Big|n^{(0)}, 1\Big\rangle = \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}}, \qquad \Big|n^{(0)}, 2\Big\rangle = \frac{|\uparrow\rangle - |\downarrow\rangle}{\sqrt{2}}. \tag{1.163}$$

They are orthonormal and obey

$$\Big\langle n^{(0)}, l\Big| V \Big|n^{(0)}, k\Big\rangle = \begin{pmatrix} v & 0 \\ 0 & -v \end{pmatrix}_{lk}. \tag{1.164}$$

We could also have identified this basis in advance by identifying what was the 'T' operator mentioned above A good choice would have been $T = S_x$: it commutes with both H_0 and V, and its two eigenvalues $\pm\hbar/2$ are distinct.

Steps 2 and 3: corrections to the states. In this model V_2 is the entire Hilbert space. There are no states outside the doublet, so

$$\Big|n^{(1)}, k\Big\rangle_{\text{out}} = 0. \tag{1.165}$$

For 'step 3', we use Eq. (1.130). The first-order energy shifts found above are

$$E^{(1)}_{n,1} = v, \qquad E^{(1)}_{n,2} = -v. \tag{1.166}$$

Thus, for $v \neq 0$, the denominators in that equation are nonzero. For the first state, the only possible inside correction is along the second state:

$$\Big|n^{(1)}, 1\Big\rangle_{\text{in}} = \frac{\big\langle n^{(0)}, 2\big| V \big|n^{(1)}, 1\big\rangle_{\text{out}}}{E^{(1)}_{n,1} - E^{(1)}_{n,2}} \Big|n^{(0)}, 2\Big\rangle \tag{1.167}$$

$$= \frac{0}{2v} \Big|n^{(0)}, 2\Big\rangle = 0. \tag{1.168}$$

However, since correction outside $\mathcal{V}_2$ was zero in this case, also the correction inside is zero. The component along the starting state itself is zero by intermediate normalization.

Finally, taking $l = k$ in Eq. (1.124) gives

$$E^{(2)}_{n,k} = \Big\langle n^{(0)}, k\Big| V \Big|n^{(1)}, k\Big\rangle_{\text{out}} = 0. \tag{1.169}$$

Consequently,

$$\boxed{\left|n^{(1)},k\right\rangle = \left|n^{(1)},k\right\rangle_{\text{out}} + \left|n^{(1)},k\right\rangle_{\text{in}} = 0, \qquad E^{(2)}_{n,k} = 0.} \tag{1.170}$$

The perturbation has already selected the symmetric and antisymmetric combinations in step 1. Once these are used as the starting states, there is no further first-order change to either state.

In fact, direct application of the full Hamiltonian gives

$$H(\lambda)\left|n^{(0)},1\right\rangle = (E^{(0)}_n + \lambda v)\left|n^{(0)},1\right\rangle, \tag{1.171}$$

$$H(\lambda)\left|n^{(0)},2\right\rangle = (E^{(0)}_n - \lambda v)\left|n^{(0)},2\right\rangle. \tag{1.172}$$

Thus the result is exact:

$$\boxed{E_{n,1}(\lambda) = E^{(0)}_n + \lambda v, \qquad E_{n,2}(\lambda) = E^{(0)}_n - \lambda v.} \tag{1.173}$$

The labels 1 and 2 refer to the symmetric and antisymmetric states; their energy ordering depends on the sign of λv.

Even an arbitrarily small nonzero coupling selects equal-weight combinations of $|\uparrow\rangle$ and $|\downarrow\rangle$. This change of starting basis is the essential step. Once that basis is chosen, no further state corrections are needed in this two-dimensional example.

1.10 Comparison of the two methods

The variational principle and perturbation theory answer different questions. The variational method asks for the best state inside a chosen family and gives a rigorous upper bound for the ground state. Perturbation theory assumes that the Hamiltonian is close to a solvable one and gives systematic corrections to energies and states.

In the variational method, the main source of error is the choice of trial space. If the trial family misses an important physical feature, such as a cusp, a node, or a long-distance tail, the upper bound may be poor. In perturbation theory, the main source of difficulty is small denominators or strong coupling between unperturbed states. A perturbation that is small in appearance can be large if it connects nearly degenerate levels.

The two methods are often combined, but their logic should not be confused. One may diagonalize a Hamiltonian in a finite variational basis and then treat a remaining weak interaction perturbatively. Conversely, perturbation theory can suggest which functions should be added to a variational basis.